

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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Test

1 Replication Target

In this project, we aim to replicate the findings of *From the French franc to the euro, is there an economic impact for the CFA franc zone countries? A 'bias-corrected' synthetic control method* by Alassane Diallo and Adama Ba, published in Cogent Economics & Finance in 2024. This paper assesses the impact of pegging the CFA franc to the euro on real GDP per capita in the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CAEMC) countries. The original study uses a set of 35 annual country-level panel data from 1980 to 2019 and applies synthetic control methods to estimate the counterfactual economic outcomes for member countries.

2 Data

Table 1: Variables

Variable	Type	Category	(First) Page Ref	Possible Source
Real GDP per capita	Continuous	Outcome	6	WDI, PennWorld
Lagged GDP per capita	Continuous	Lagged outcome	10	WDI, PennWorld
Gross Fixed Capital Formation (% GDP)	Continuous	Capital	10	
Government expenditure (% GDP)	Continuous	Fiscal	10	WDI, PennWorld
Official Development Assistance (% GDP)	Continuous	External	10	Source not listed
Foreign Development Investment (% GDP)	Continuous	External	10	WDI
Employment rate	Continuous	Labour	10	WDI
Agriculture (% GDP)	Continuous	Structure	10	WDI, PennWorld
Industry (% GDP)	Continuous	Structure	10	WDI, PennWorld
Governance / institutional quality	Index	Institutions	10	Polity5 Dataset
Political regime characteristics	Categorical/Index	Institutions	10	Polity5 Dataset
Inflation	Continuous	Macro controls	5	WDI, PennWorld
Euro peg dummy	Binary	Treatment		Authors
Synthetic GDP per capita	$\tilde{Y}_{it}$	Continuous	Constructed	
Treatment effect	τ_{it}	Continuous	Constructed	

Table 2: CFA franc zone countries annual inflation rate: WAEMU

country	1980–2001	2002–2021
Benin	7.5270	2.1268
Burkina Faso	4.3427	2.2660
Cote d’Ivoire	8.5558	2.1842
Guinea-Bissau	39.2343	2.8921
Mali	8.6642	3.5490
Niger	6.0235	2.8630
Senegal	5.1354	1.8448
Togo	6.2716	2.0069
Average inflation for the WAEMU zone	10.7193	2.4666
Average inflation without Guinea-Bissau	6.6457	2.4058

Table 3: CFA franc zone countries annual inflation rate: CAEMC

country	1980–2001	2002–2021
Cameroon	7.0840	1.8268
Central African Republic	6.5481	2.9430
Chad	5.2621	5.4660
Congo, Rep.	7.3655	6.1262
Equatorial Guinea	11.0392	6.7586
Gabon	7.4425	3.7486
Average inflation for the CAEMC zone	7.4569	4.4782

Table 4: Non-CFA franc zone countries mean annual inflation rate

country	1980–2001	2002–2021
Angola	550.9314	32.1383
Burundi	10.3255	9.2785
Congo, Dem. Rep.	1810.0588	16.3427
Ethiopia	5.2079	13.5134
Gambia, The	14.5303	10.1870
Ghana	36.1786	21.0285
Guinea	16.1342	11.2597
Kenya	11.8579	8.5152
Madagascar	18.4232	8.3332
Malawi	23.6956	18.1634
Nigeria	30.4076	13.5749
Sao Tome and Principe	31.6514	10.6113
Sierra Leone	51.3708	11.1235
Sudan	50.6410	37.8117
Tanzania	21.8489	7.4472
Uganda	53.1089	8.7408
Zambia	51.1202	12.4355
Zimbabwe	-1.6381	71.6617
Average inflation	154.7697	17.8981

Table 5: WAEMU countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Benin	1.0235	1.0352	1.0665	1.0540
Burkina Faso	1.0213	1.0213	1.0066	1.0678
Cote d'Ivoire	0.9847	1.0481	1.0273	1.0522
Guinea-Bissau	1.0112	1.0037	0.9620	1.0572
Mali	1.0615	1.0303	1.0187	1.0586
Niger	0.9934	1.0067	0.9991	1.0587
Senegal	0.9983	1.0078	0.9863	1.0502
Togo	0.9903	1.0427	0.9923	1.0481
Average for the WAEMU area	1.0105	1.0245	1.0074	1.0559
Average without Guinea-Bissau	1.0104	1.0274	1.0138	1.0557

Table 6: CAEMC countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Cameroon	0.9961	1.0178	0.9911	1.0467
Central African Republic	0.9868	1.0025	0.9609	1.0432
Chad	0.9908	1.0347	0.9916	1.1043
Congo, Rep.	1.0200	1.0522	1.0029	1.0729
Equatorial Guinea	1.2073	1.0537	1.2602	1.1028
Gabon	1.0071	1.0372	1.0011	1.0490
Average for the CAEMC zone	1.0347	1.0330	1.0346	1.0698

Table 7: Non-CFA countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Angola	0.9892	1.0757	0.9813	1.1195
Burundi	0.9770	1.0039	0.9675	1.0379
Congo, Dem. Rep.	0.9078	1.0381	1.1332	1.0809
Ethiopia	0.9954	1.0839	0.9449	1.1115
Gambia, The	1.0012	0.9921	1.0723	1.0453
Ghana	1.0087	1.0323	0.9871	1.1360
Guinea	0.9770	0.9955	0.9886	1.0558
Kenya	0.9972	1.0383	1.0209	1.0868
Madagascar	0.9942	1.0106	1.0232	1.0281
Malawi	0.9803	1.0229	1.0217	1.0738
Nigeria	1.0205	1.1040	1.0919	1.0981
Sao Tome and Principe	1.0179	1.0284	0.9814	1.0823
Sierra Leone	0.9870	1.0315	1.0978	1.0515
Sudan	1.0116	1.0402	1.0677	1.0388
Tanzania	1.0144	1.0372	1.0406	1.0570
Uganda	1.0358	1.0264	0.9921	1.0776
Zambia	0.9996	1.0526	0.9827	1.0701
Zimbabwe	0.9869	1.0036	0.9753	1.1332
Average	0.9945	1.0343	1.0206	1.0769

Table 8: WAEMU countries' mean annual GDP per capita growth: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Benin	2.346	3.5210	1.4790	1.4790
Burkina Faso	2.128	2.1310	2.6980	2.6500
Cote d'Ivoire	-1.531	4.8070	-1.1260	1.4570
Guinea-Bissau	1.117	0.3740	0.7170	1.1300
Mali	6.151	3.0350	2.5530	0.6750
Niger	-0.662	0.6730	-1.1750	1.2940
Senegal	-0.167	0.7800	0.7000	1.4350
Togo	-0.971	4.2710	-0.1760	1.1360
Average for the WAEMU area	1.051	2.4490	0.7090	1.4070
Average without Guinea-Bissau	1.042	2.7454	0.7076	1.4466

Table 9: CAEMC countries' mean annual GDP per capita growth: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Cameroon	-0.394	1.778	-1.273	1.066
Central African Republic	-1.317	0.249	-1.278	-0.080
Chad	-0.920	3.472	-0.203	2.316
Congo, Rep.	2.000	5.218	-1.203	-0.795
Equatorial Guinea	20.730	5.369	33.076	0.213
Gabon	0.710	3.720	-0.812	-0.728
Average for the CAEMC zone	3.468	3.301	4.718	0.332

Table 10: Non-CFA countries' mean annual GDP per capita growth: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Angola	-1.080	7.569	-1.293	1.432
Burundi	-2.298	0.394	-3.034	0.049
Congo, Dem. Rep.	-9.217	3.809	-8.169	2.108
Ethiopia	-0.422	8.390	0.003	5.781
Gambia, The	0.116	-0.795	0.305	0.157
Ghana	0.871	3.227	1.746	3.462
Guinea	-2.300	-0.454	1.252	2.247
Kenya	-0.277	3.831	-0.895	1.768
Madagascar	-0.579	1.062	-1.023	-0.393
Malawi	-1.973	2.291	1.134	1.959
Nigeria	2.047	10.405	-0.632	2.152
Sao Tome and Principe	1.790	2.839	0.060	1.953
Sierra Leone	-1.305	3.146	-3.661	2.642
Sudan	1.158	3.806	3.862	-0.546
Tanzania	1.438	3.720	0.538	3.101
Uganda	3.577	2.640	3.166	2.919
Zambia	-0.045	5.263	-0.514	2.231
Zimbabwe	-1.315	0.359	0.346	-0.771
Average	-0.545	3.417	-0.378	1.792